Magic VLSI - theory
part-1

$ magic -T scmos
$ Ctrl + Z to kill
$ magic -T scmos& (to run in the background)
$ magic -T scmos filename.mag & (to open a preexisting file)

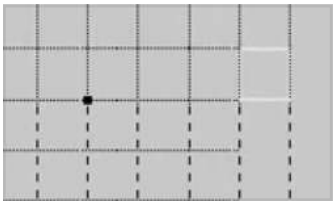
- Magic has two windows - command window and canvas
- Enterring commands -

1. Magic command window -

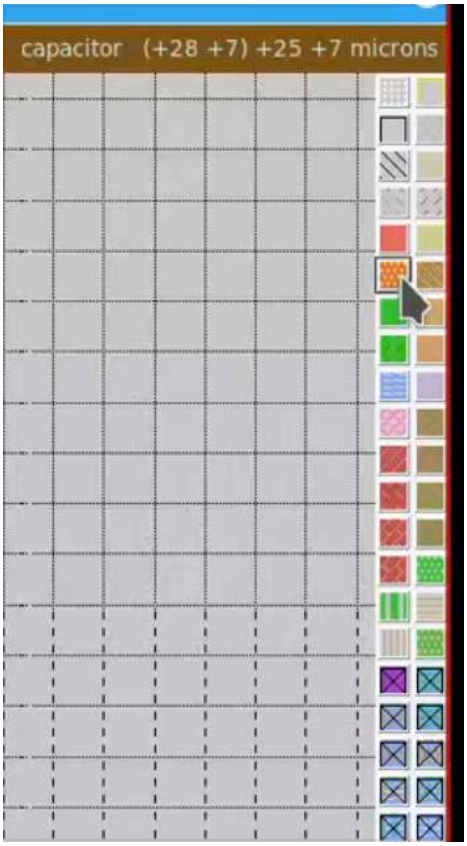
% grid on
Alternate : Keep the four directional cursor on the layout + press G
Z - for zoom out
Shift + z - zoom in

2. Second way of entering commands is by keeping the canvas/layout window active and pressing colon (:) in the command window

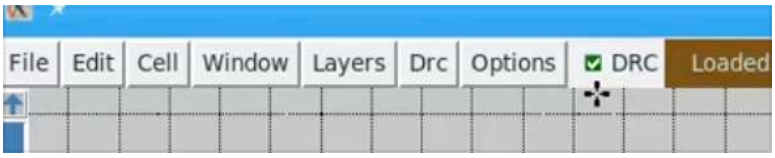
- Options > Tech manager > scmos
- To close the command window -: exit
- General info -



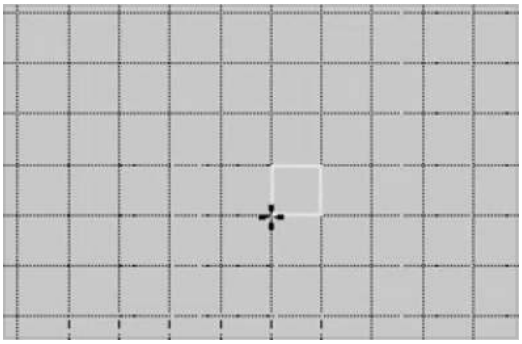
Black dot at (0,0)



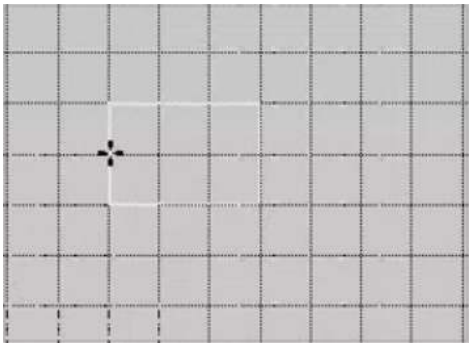
RHS : pallets (switch it ON/OFF in Options > Toolbar)



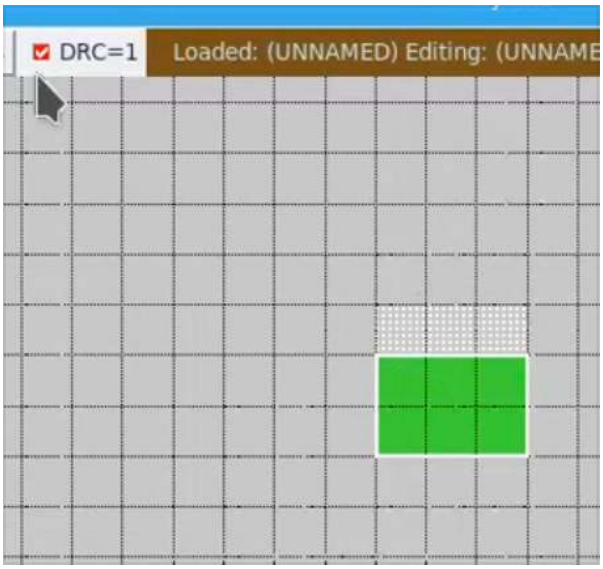
DRC - Design Rule Check (Green - OK + proceed)
Check for minimum dimensions and spacing between geometrical objects



Cursor + left click-> anchors that particular spot
Cursor + right click after above step -> to select given area
Example - 3x2 dimension



Paint the selected area by choosing from pallete
Ex: green - for n diffusion
Alternate: place the cursor in the selected area, go to command window and press ' : '
: paint ndiff



Gray area corresponds to the substrate (P)
Top view is visible on the canvas
Dotted rectangular area = error (also notice red check next DRC - 1 DRC violation)
: drc find

```
texthelper updatedisplay uplevel upsidedown upvar
Unknown macro or short command: 'XK_q'
: paint ndiff
: drc find
Error area #1:
N-type Diffusion width must be at least 3 (MOSIS rule #2.1a)
```

To remove any layer, pick that area (white outline) and click anywhere on the substrate.

To move a selected area:

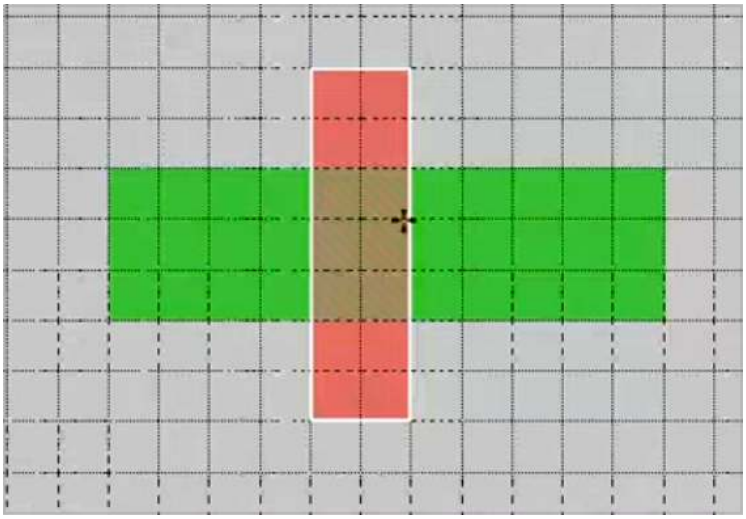
Click on it -> gets white outline
: select area
Now the outline can be freely moved by using command window-
: move e 1
(move east by 1 unit)
Press ' . ' to run the same command again

To get specific help with commands -
% help (general)
% help move (will give all move related commands)
% help select
% select help

- Designing NMOS -

After paint ndiff

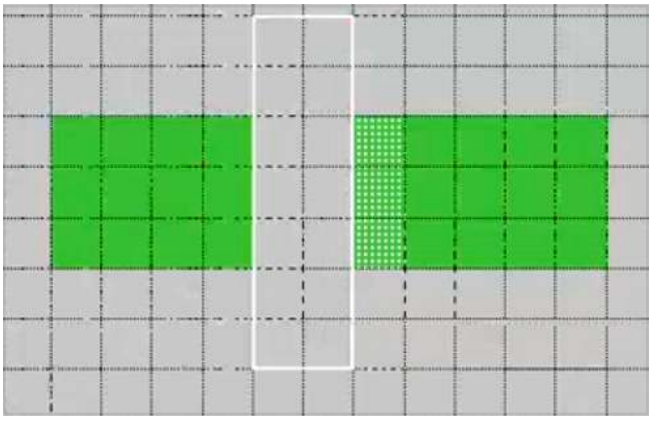
Poly -
: paint poly



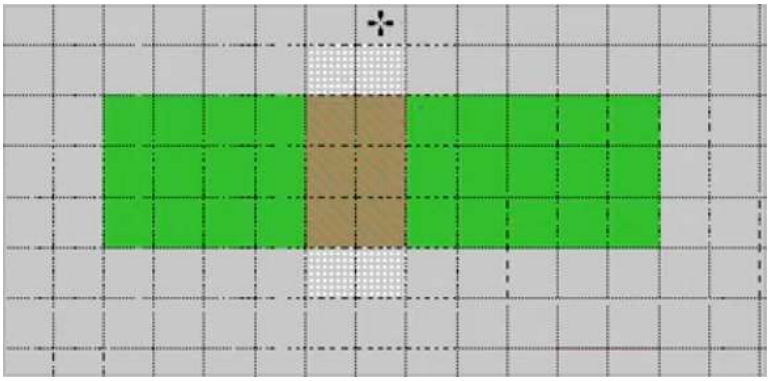
Pure red - poly overhang
Green + red - gate

- To selectively only remove poly -

: select area poly
(syntax - select area layer)
: erase



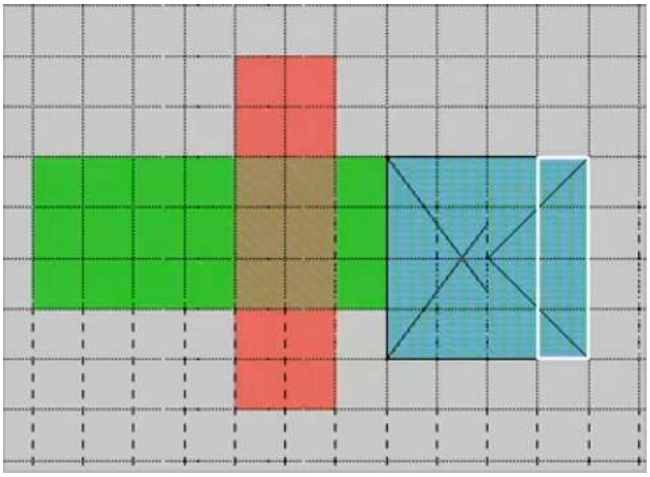
Method 2 : Instead of erase -
: delete (after selecting)

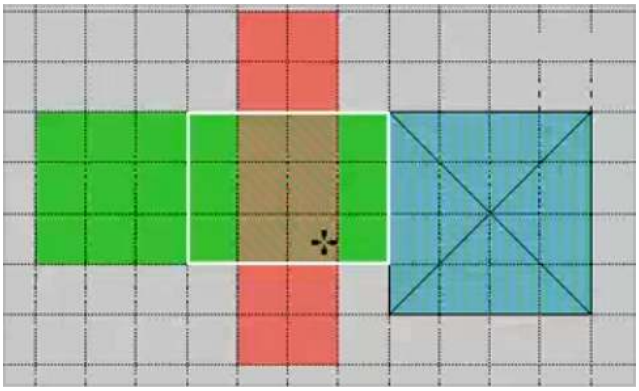


Gate area still remains
Now to remove this -
: select area nfet
: delete

Note - width of nmosfet (green) = 3 grids = 3*lambda
Length = 2 grids = 2*lambda (p.s. the remaining green area is Source and drain)

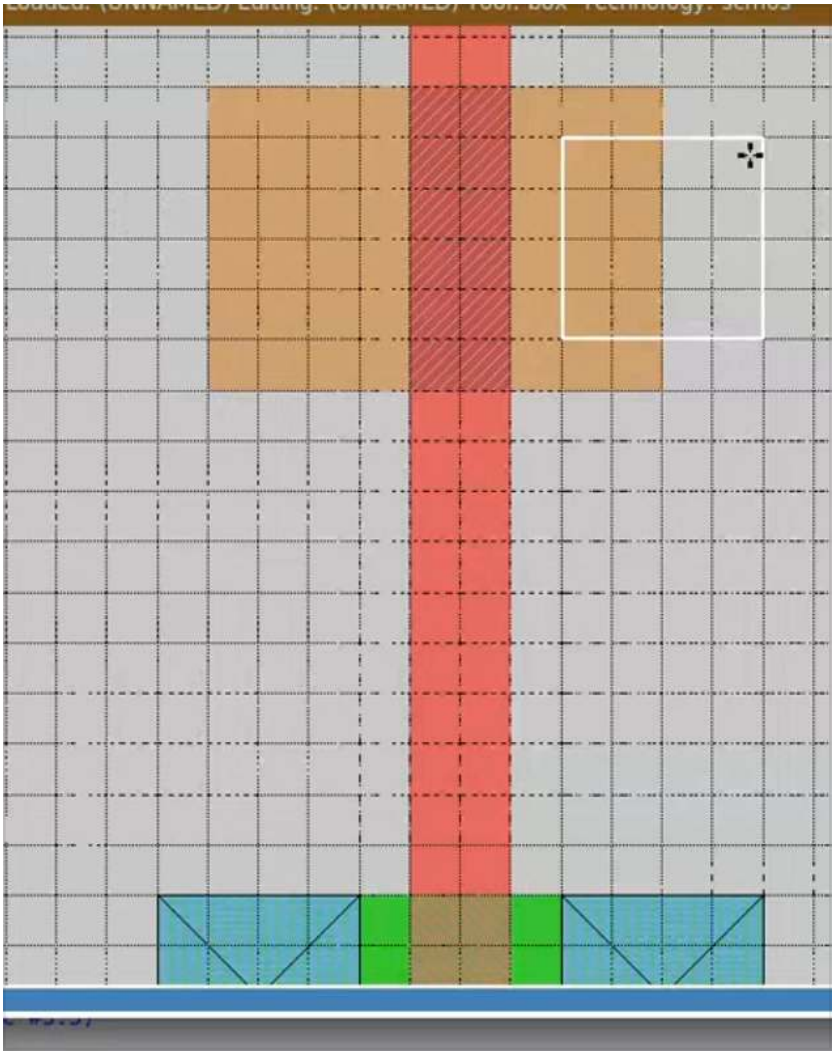
Add contacts -
: paint ndc
(N diffusion contact)



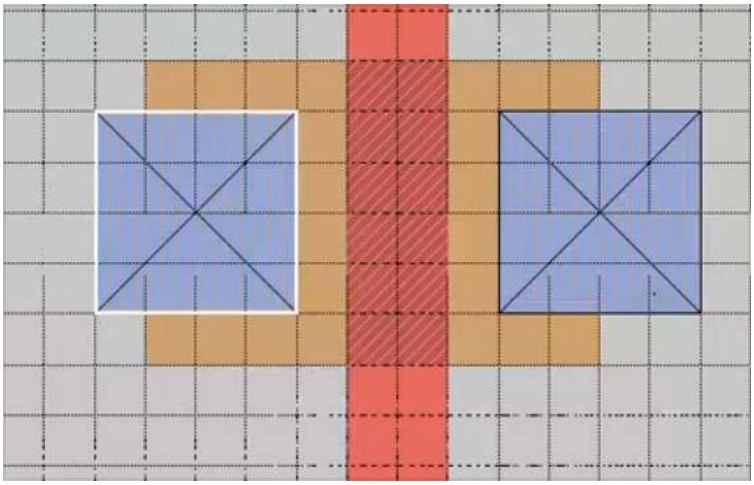


Replicate contact on lhs

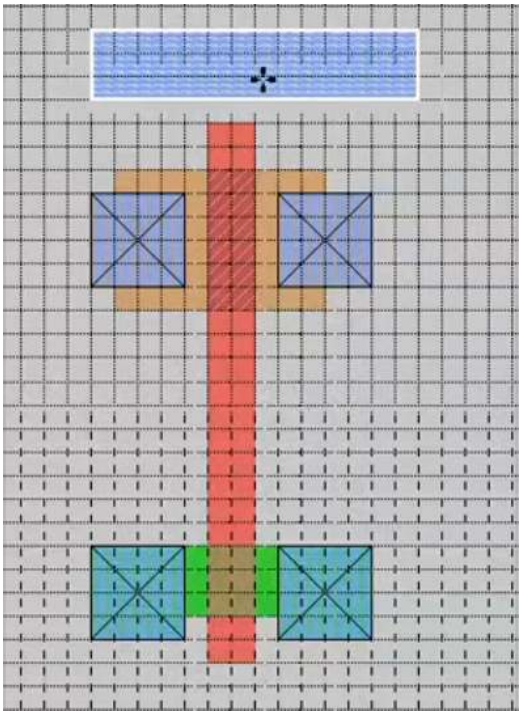
PMOS - diff has to be twice the size of ndiff



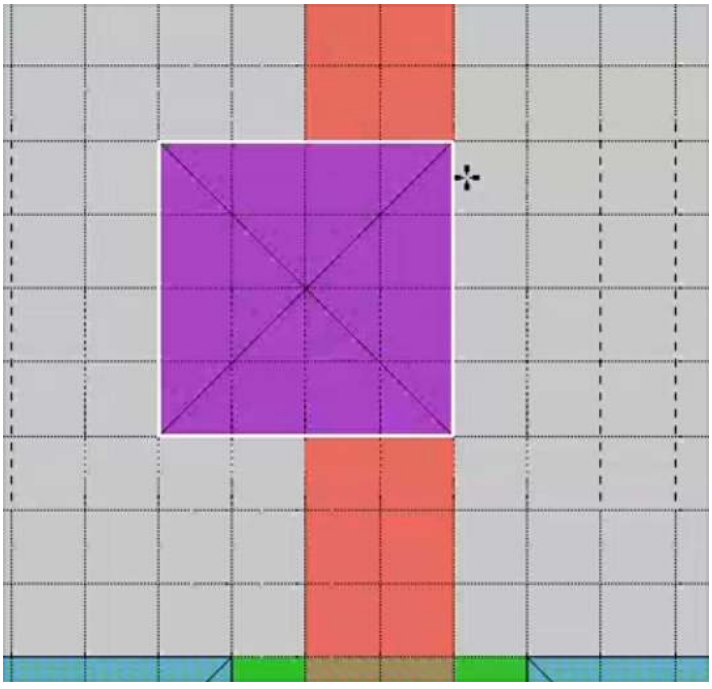
: paint pdc



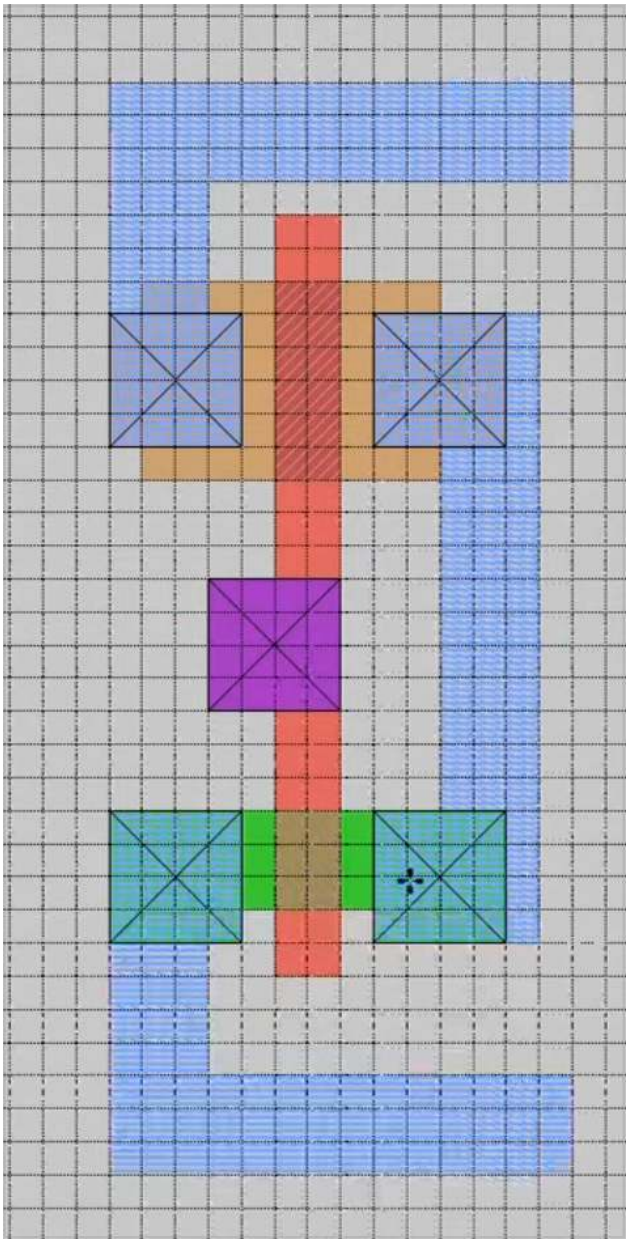
: paint m1
(Metal 1)



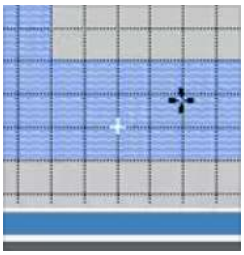
: paint pc
(Poly contact)



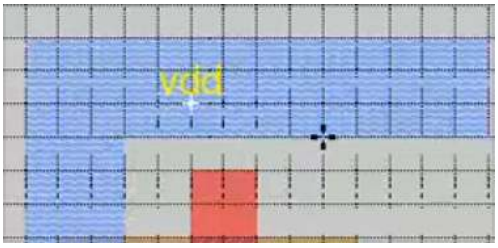
Inverter -



Labelling -
Click a point on the area, then in command line -



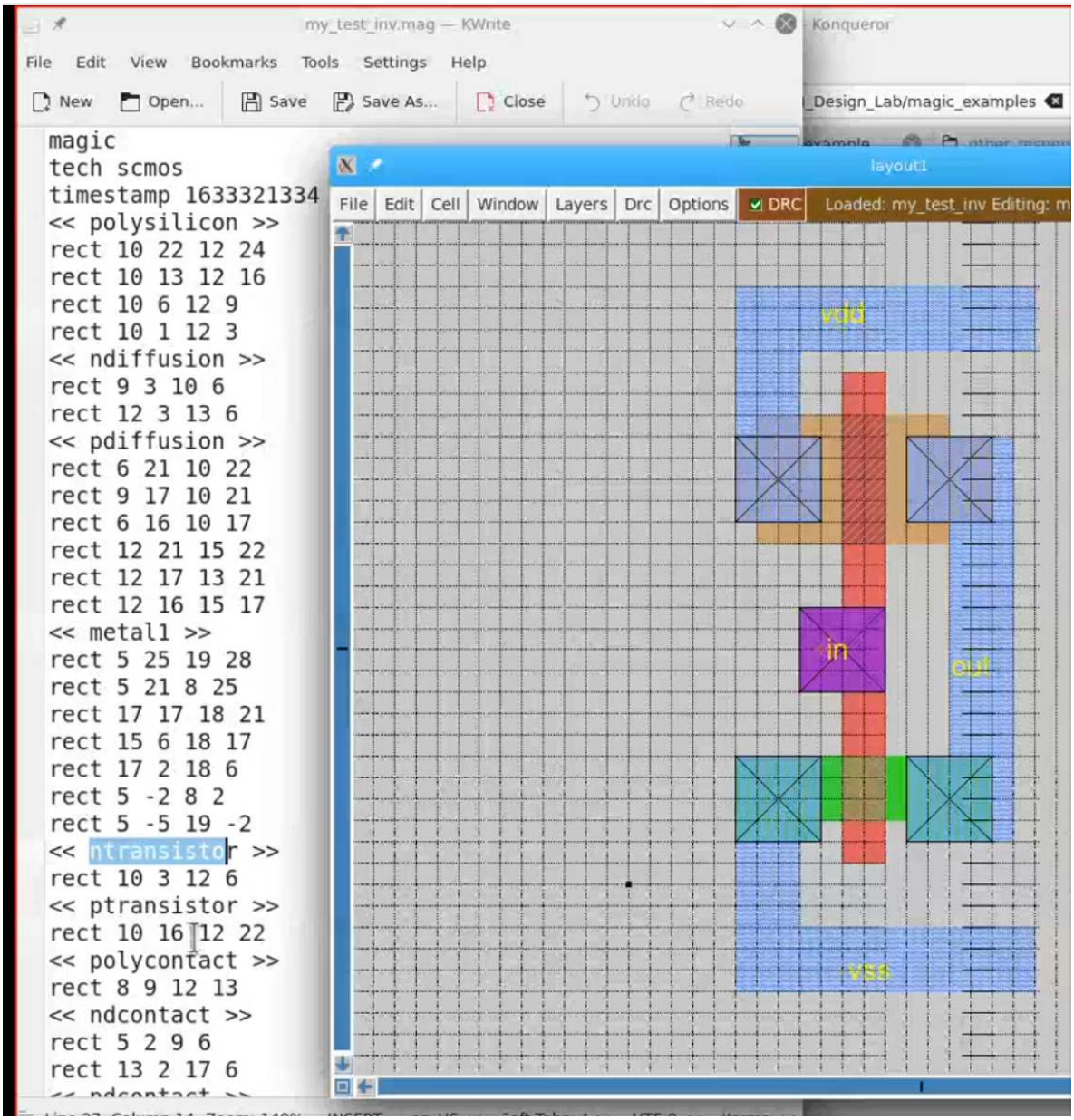
Syntax => :label str [pos [layer]]
:label vdd n
(n - north of point selected)



- Saving the file -

: save filename.mag

part2



Converting these rect components to electrical components -

% extract

```
%
% extract
Extracting my_test_inv into my_test_inv.ext
```

my_test_inv.ext — KWrite

File Edit View Bookmarks Tools Settings Help

New Open... Save Save As... Close Undo Redo

```
timestamp 1633321334
version 8.3
tech scmos
style lambda=1.0(scna20_orb)
scale 1000 1 100
resistclasses 26670 59550 23860 19690 27260 2000000 49 26
250583(
node "vss" 45 2538 5 2 ndc 19 18 0 0 0 0 0 0 0 0 0 0 70 52
0 0 0 0
node "out" 176 1927 12 3 ndiff 19 18 26 22 0 0 0 0 0 0 0 0
73 52 0 0 0 0
node "vdd" 105 2444 5 17 pdc 0 0 28 22 0 0 0 0 0 0 0 0 70
52 0 0 0 0
node "in" 272 4924 8 9 pc 0 0 0 0 54 54 0 0 0 0 0 0 16 16 0
0 0 0
device mosfet nfet 10 3 11 4 2 3 "Gnd!" "in" 4 0 "vss" 3 0
"out" 3 0
device mosfet pfet 10 16 11 17 2 6 "Vdd!" "in" 4 0 "vdd" 6
0 "out" 6 0
subcap "out" -1410
```

Ext file to spice netlist -
% ext2spice

my_test_inv.spice — KWrite

File Edit View Bookmarks Tools Settings Help

New Open... Save Save As... Close Undo Redo

```
* SPICE3 file created from my_test_inv.ext -
technology: scmos

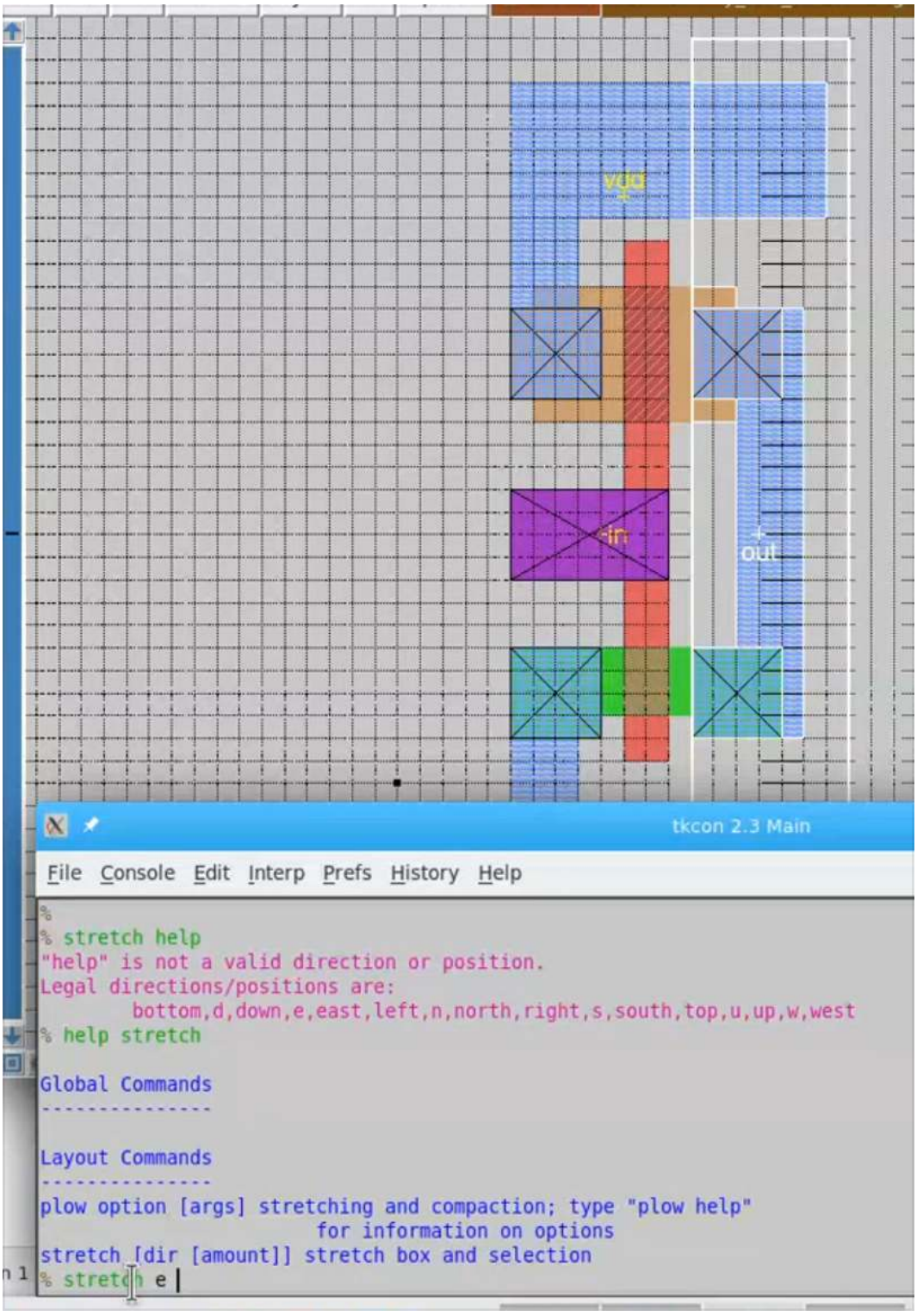
.option scale=1u

M1000 out in vss Gnd nfet w=3 l=2
+  ad=0 pd=0 as=0 ps=0
M1001 out in vdd Vdd pfet w=6 l=2
+  ad=0 pd=0 as=0 ps=0
C0 vss Gnd 2.54fF
C1 vdd Gnd 2.44fF
C2 in Gnd 4.92fF
```

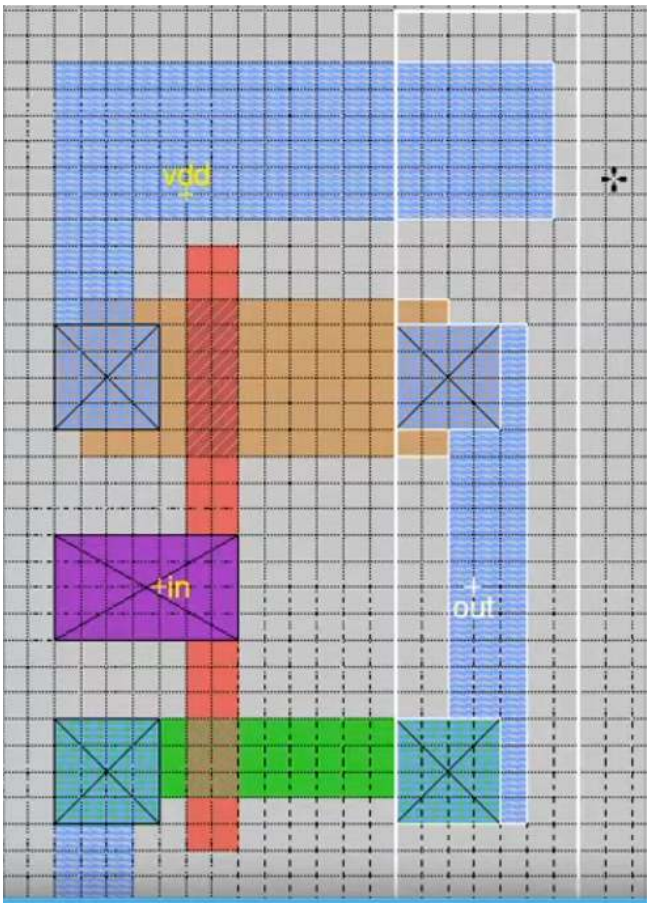
Changing M1 and M2 will affect the capacitances-



Stretching without disconnecting -



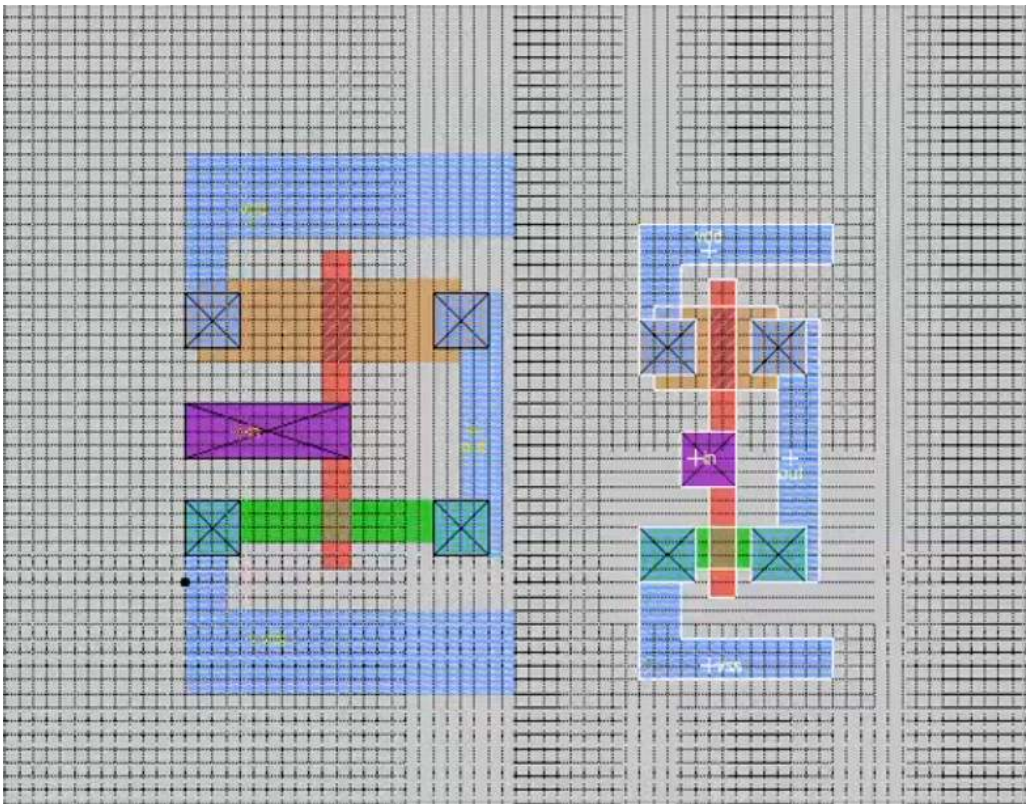
: select area
: stretch e 5
Result -



To get rid of that white outline -
: select clear

To select multiple objects -
: select more area

To copy-paste the cell (in row fashion) -
: dump filename.mag



Original spice netlist extracted from .mag file -

New Open... Save Save As... Close Undo

```
* SPICE3 file created from my_test_inv.ext - technology: scmos

.option scale=1u

M1000 out in vss Gnd nfet w=3 l=2
+  ad=0 pd=0 as=0 ps=0
M1001 out in vdd Vdd pfet w=6 l=2
+  ad=0 pd=0 as=0 ps=0
C0 vss Gnd 2.54fF
C1 vdd Gnd 2.44fF
C2 in Gnd 4.92fF
```

Inorder to simulate this -
Make the following changes (note - .include <technology file) -

```
* SPICE3 file created from my_test_inv.ext - technology: scmos

.option scale=1u

.include .....

M1000 out in vss Gnd nfet w=3 l=2
+  ad=0 pd=0 as=0 ps=0
M1001 out in vdd Vdd pfet w=6 l=2
+  ad=0 pd=0 as=0 ps=0
C0 vss Gnd 2.54fF
C1 vdd Gnd 2.44fF
C2 in Gnd 4.92fF

V_DD vdd 0 5
V_SS vss 0 0
vin in 0 5

dc 0 5 0.1
```

And so on.. complete the netlist and plot accordingly.

Look into command extres (extract parasitic resistance?)

To compare layout Vs schematic - use **NETGEN** (LVS comparison)

For project use NETGEN and IRSIM

: extract
% ext2sim

Sim file (converts entire layout to switches and the corresponding parasitic RCs in that) -

```
| units: 100 tech: scmos format: MIT
n in vss out 2 3 10 3
p in vdd out 2 6 10 16
C vss GND 2.54
R vss 45
R out 176
C vdd GND 2.44
R vdd 105
C in GND 4.92
R in 272
```

Simulate this using IRSIM

For project, evaluation parameters include - area, speed, energy per transition